Notes: Grubb & Osborne (AER, 2015)

Cellular Service Demand: Biased Beliefs, Learning, and Bill Shock

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1 Big picture

- **Question.** How should economists model cellular-phone service demand when consumers choose nonlinear contracts before knowing their future usage, and what are the welfare effects of bill-shock regulation?
- **Setting.** The paper uses a detailed panel of individual cellular bills from 2002–2004. The plans are three-part tariffs: a monthly fee, an allowance of included minutes, and an overage price for use above the allowance.
- **Behavioral ingredients.** Consumers are uncertain about future calling tastes, learn from past usage, are inattentive to past usage when making some choices, and are overconfident about future usage risk.
- **Key empirical facts.** Usage responds sharply to marginal prices, consumers do not bunch at allowances, and consumers keep responding to price changes even when far below allowance.
- **Core demand model.** Consumers first choose a plan, then choose calling thresholds for peak and off-peak calls, and then realize tastes and make calls. Plan choice depends on beliefs about future taste distributions.
- **Core estimates.** The model estimates strong price sensitivity and strong overconfidence. The overconfidence parameter is about $\delta = 0.38$, meaning perceived uncertainty is far smaller than true uncertainty.
- **Biases.** The paper estimates three biases: overconfidence, aggregate mean bias, and conditional mean bias. Overconfidence makes consumers choose risky plans; aggregate mean bias pushes consumers toward small plans; conditional mean bias pushes them toward extreme plans.
- **Policy result.** Bill-shock alerts can mechanically reduce overage charges, but after firms re-optimize tariffs, alerts lower average annual consumer welfare by about 33 per consumer in the sample environment.
- **Why policy can backfire.** When regulation reduces overage revenue, firms adjust prices by raising fixed fees and changing plan menus. Consumers call less efficiently, and cross-subsidies embedded in overage pricing change.
- **Main takeaway.** Consumer protection policies that reveal usage information can have very different welfare effects under fixed prices and under equilibrium price adjustment.

2 Incremental contribution of the paper

2.1 Contribution relative to standard nonlinear pricing models

Standard nonlinear pricing models often assume that consumers know their demand perfectly when they choose plans. This paper shows that such models fit cellular usage poorly. Consumers do not simply choose a precise quantity and then minimize cost across plans. Instead, they face uncertainty about future usage and choose plans under imperfect beliefs. The paper therefore shifts the analysis from deterministic tariff choice to stochastic plan choice with learning and biased beliefs.

2.2 Contribution relative to behavioral industrial organization

The paper gives a structural behavioral model in a market with rich nonlinear contracts. The behavioral biases are not simply assumed; they are disciplined by usage patterns, plan choices, and post-choice usage histories. Overconfidence is identified by systematic choices of risky plans, while learning is identified by dynamics in plan choices and usage after experience. This turns a behavioral explanation of bill shock into a quantitatively disciplined demand system.

2.3 Contribution relative to policy analysis of bill-shock regulation

The paper goes beyond documenting that consumers make mistakes. It evaluates a policy that was salient to regulators: alerts when consumers approach or exceed their usage allowance. The central contribution is to

distinguish fixed-price effects from endogenous-price effects. Fixed-price calculations make bill-shock alerts appear consumer friendly, but equilibrium price adjustment reverses the sign of consumer surplus.

2.4 Contribution relative to empirical work on consumer inattention

The paper uses detailed consumption timing within billing cycles to identify inattention and belief errors. Usage jumps at the start of free night/weekend periods even among consumers far below allowance, showing that consumers react to expected marginal prices rather than perfectly tracking their final bill position. This evidence ties micro usage behavior directly to the informational policy question.

3 Data and institutional setting

3.1 Cellular plans and three-part tariffs

Consumers choose among plans that specify:

- a monthly fixed fee M_j;
- a quantity allowance Q_j of included minutes;
- an overage price p_j per billable minute above Q_j.

The four popular plans in the sample are labelled plan 0 through plan 3. Plan 0 is close to a flat-rate/no-allowance option with a low monthly fee and a positive per-minute price. Plans 1–3 offer higher monthly fees and larger allowances.

Interpretation: The pricing menu generates risk. A larger plan provides insurance against high usage but wastes money if usage is low. Overconfidence is revealed by consumers choosing too little insurance against high usage risk.

3.2 Billing and usage categories

The paper distinguishes peak and off-peak calls. Peak minutes are the main source of overage risk. Off-peak usage is often free or priced differently because many plans include free nights and weekends. The model suppresses some in-network/out-of-network distinctions in the main text but handles them in an appendix.

Interpretation: Separating peak and off-peak usage matters because price sensitivity is identified by sharp changes in calling at the moment off-peak pricing begins.

3.3 Savings opportunities

A consumer has a savings opportunity when her realized usage and chosen plan imply that she could have paid less under another plan. The paper measures savings opportunities across enrollment periods and plan switches.

Interpretation: Savings opportunities are descriptive evidence of plan-choice mistakes. They are not sufficient for welfare analysis because ex ante risk and ex post mistakes must be distinguished.

3.4 Stylized facts from usage behavior

The paper documents three facts:

1. consumers are price sensitive;
2. consumers choose usage under uncertainty rather than with perfect foresight;
3. consumers are inattentive to accumulated usage within the billing cycle.

Interpretation: These facts motivate a model in which consumers choose thresholds, not exact quantities, and in which plan choice depends on biased beliefs about future usage volatility.

4 Model

4.1 Billing quantity and tariff

For consumer i, plan j, and period t, billable minutes are

$$q_{ijt}^{billable} = q_{ijt}^{pk} + OP_j q_{ijt}^{op}, \tag{1}$$

where OP_j indicates whether plan j charges off-peak usage.

The payment under plan j is

$$P_j(q_{ijt}) = M_j + p_j \max\left\{0, q_{ijt}^{billable} - Q_j\right\}. \tag{2}$$

Interpretation: The tariff is nonlinear. Consumers face zero marginal price below allowance and a positive marginal price above allowance, but they do not know future usage when choosing the plan.

4.2 Monthly utility

Consumer utility in month t from plan j is

$$u_{ijt} = \sum_{k \in \{pk, op\}} V(q_{ijt}^k, \theta_{it}^k) - P_j(q_{ijt}) + \eta_{itj}. \tag{3}$$

The calling value function is

$$V(q_{ijt}^k, \theta_{it}^k) = \frac{1}{\beta} q_{ijt}^k \left(1 - \frac{1}{2} \frac{q_{ijt}^k}{\theta_{it}^k}\right). \tag{4}$$

Interpretation: The parameter β controls price sensitivity. The taste shock θ shifts the inverse demand curve and thus the desired amount of calling.

4.3 Timing

At each period:

1. the consumer receives a signal about current tastes and updates beliefs;
2. she chooses a plan from the available menu;
3. she chooses calling thresholds conditional on plan prices and beliefs;
4. tastes are realized and actual usage occurs;
5. beliefs are updated using realized usage.

Interpretation: Plan choice and calling are separated. A consumer can choose a plan under uncertainty and later make calls once tastes are realized.

4.4 Calling threshold

The optimal threshold v_{itj}^{pk} is characterized by

$$v_{itj}^{pk} = p_j \Pr\left(\theta_{it}^{pk} \geq Q_j / \hat{q}(v_{itj}^{pk}) \mid \Im_{it}\right) \frac{E\left[\theta_{it}^{pk} \mid \theta_{it}^{pk} \geq Q_j / \hat{q}(v_{itj}^{pk}), \Im_{it}\right]}{E\left[\theta_{it}^{pk} \mid \Im_{it}\right]}. \tag{5}$$

Interpretation: The threshold lies between zero and the overage price. It rises when the consumer thinks she is more likely to exceed allowance or when conditional overage tastes are high.

4.5 Beliefs and learning

Consumers are uncertain about future taste shocks. Beliefs combine population heterogeneity, individual persistent components, and month-specific shocks. Consumers update beliefs using observed past usage, but the model allows biased beliefs.

The key bias parameters are:

- δ: overconfidence; lower δ means consumers perceive too little variance;
- b_1: aggregate mean bias; negative estimates imply consumers underestimate average usage;
- b_2: conditional mean bias; positive estimates imply consumers choose plans that are too extreme.

Interpretation: The model decomposes mistakes into risk misperception and mean errors. This matters because different biases have different implications for plan choice and regulation.

5 Identification and estimation

5.1 Price sensitivity

Price sensitivity is identified by usage responses around times when the marginal price changes sharply, especially the beginning of free nights and weekends. If usage rises exactly when calls become free, consumers are responding to marginal prices.

Interpretation: This variation identifies calling demand without relying only on cross-sectional plan choices.

5.2 Usage uncertainty

Lack of bunching at allowance thresholds indicates that consumers cannot perfectly foresee usage. If they knew future usage precisely, many consumers would stop just below allowance when overage prices are high.

Interpretation: The smooth usage density around allowances supports a stochastic usage model rather than a deterministic quantity-choice model.

5.3 Inattention

Consumers increase calling when free periods begin even in months when final usage is far below allowance. This shows that consumers are not fully conditioning on the exact remaining balance of minutes in the billing cycle.

Interpretation: Bill-shock alerts can be informative because consumers do not perfectly track usage histories.

5.4 Plan choices and beliefs

Plan choice shares identify beliefs about usage distributions. Consumers choosing overly risky plans identify overconfidence. Changes in plan choices over time identify learning. The model also uses price variation over time to separate means and variances of beliefs.

Interpretation: Beliefs are identified by the way consumers sort across the nonlinear tariff menu. In particular, choosing a plan with too little allowance reveals underestimation of usage risk.

5.5 Estimation strategy

The paper estimates demand parameters using observed plan choices, usage outcomes, and plan menus. The likelihood combines plan choice probabilities and usage densities. Counterfactual simulations use estimated demand and a simple model of oligopoly price adjustment.

Interpretation: The structural approach is necessary because welfare depends on unobserved beliefs and on how firms change tariffs in response to regulation.

6 Counterfactuals and welfare

6.1 Removing biased beliefs with fixed prices

Holding observed prices fixed, the paper simulates the effect of removing biases. Removing overconfidence or all biases raises consumer surplus and reduces firm profits. Consumers switch toward safer plans and pay fewer overage charges.

Interpretation: At fixed prices, consumer mistakes transfer surplus from consumers to firms through overage payments and inefficient plan choices.

6.2 Bill-shock regulation with fixed prices

Bill-shock alerts eliminate or reduce surprise overages by informing consumers when usage approaches an allowance. At fixed prices, alerts reduce overage fees and can increase consumer surplus.

Interpretation: A fixed-price analysis makes the regulation look attractive because it prevents costly mistakes without changing the tariff menu.

6.3 Bill-shock regulation with endogenous prices

When firms re-optimize prices, the welfare effect changes. Firms reduce reliance on overage fees, adjust allowances and fixed fees, and alter the plan menu. The paper predicts that bill-shock regulation lowers annual consumer welfare by about 33 per consumer.

Interpretation: Endogenous pricing is the key policy mechanism. Regulation changes the form of nonlinear pricing, not just consumer information.

6.4 Distributional consequences

Bill-shock regulation creates winners and losers. Some consumers benefit from avoiding large overages, but others face higher monthly fees and reduced calling value. The distribution of surplus changes is skewed, with a tail of winners but a negative average.

Interpretation: Policymakers may value reductions in large bills even if average consumer surplus falls, but the paper shows the efficiency cost of that insurance-like regulation.

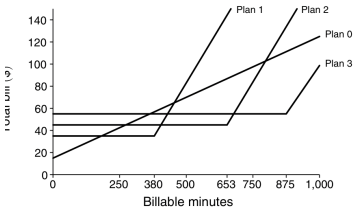
7 Tables and figures

7.1 Figure 1 and Table 1: plan menu and savings opportunities

Read:

- Figure 1 shows the four popular plans and their nonlinear pricing schedules.
- Higher plans have higher monthly fees and larger allowances.
- Table 1 reports cases where consumers could have saved money by choosing a different plan.
- Affected consumers could save from about 2.68 to 8.73 per affected bill depending on enrollment period and plan switch.

Economic message: The tariff menu creates a risk-insurance tradeoff. Savings opportunities show that many consumers choose plans that are ex post dominated, motivating a model of uncertainty and biased beliefs.



Plan	M_j	Q_j	p_j
Plan 0	$14.99	0	$0.11
Plan 1	$34.99	380	$0.45
Plan 2	$44.99	653	$0.40
Plan 3	$54.99	875	$0.40

FIGURE 1. POPULAR PLAN PRICES, SPRING 2003

egardless of any future promotions or discounts, until they switch plans or termi-

Figure 1: Popular plan prices

TABLE 1—SAVINGS OPPORTUNITIES

Opportunity	(1)	(2)	(3)
Enrollment dates	10/02–8/03	9/03 onward	10/02–8/03
Enrollment change	plan 1–3 → plan 0	plan 1 → plan 2	plan 1 → plan 2
Affected customers	246 (34 percent)	437 (56 percent)	96 (14 percent)
Savings per affected bill	$8.73	$2.68	$5.45

Notes: Savings opportunities indicate that consumers choose overly risky plans (overconfidence) predictably. Savings estimates are a lower bound because we cannot always distinguish in- and out-of-network calls.

Table 1: Savings opportunities

7.2 Figures 2 and 3: price sensitivity and usage uncertainty

Read:

- Figure 2 shows sharp increases in usage when night or weekend free periods begin.
- The response is strongest in periods where marginal prices fall to zero.
- Figure 3 shows usage densities for popular plans.
- Usage does not bunch strongly at allowances, even though overage prices are high.

Economic message: Consumers are price sensitive, but they do not know future usage precisely. Both facts are essential for the model.

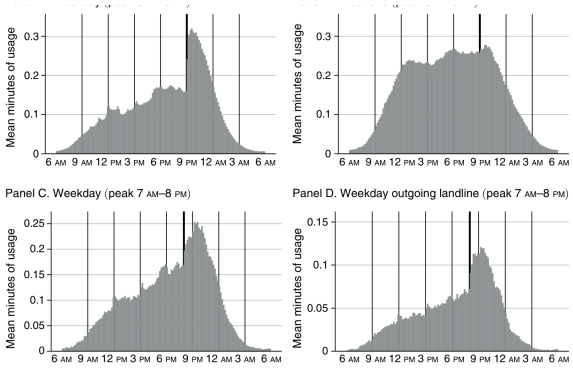


FIGURE 2. DAILY USAGE PATTERNS FOR SUBSCRIBERS WITH FREE NIGHTS AND WEEKENDS

Notes: Top row: Weekday (panel A) and weekend (panel B) usage patterns for subscribers with 6 AM–9 PM peak hours. Bold vertical lines mark 9 PM. Bottom row: Weekday usage patterns for subscribers with 7 AM–8 PM peak hours. Bold vertical lines mark 8 PM. Panel C shows all weekday calling, while panel D is restricted to outgoing calls to landlines.

nights-and-weekends. For plans with free weeknight calling starting at 8 PM, there is still a secondary increase in usage at 9 PM (panel C of Figure 2). Restricting attention to outgoing calls made to landlines (recipients for whom the cost of receiving

Figure 2: Daily usage patterns

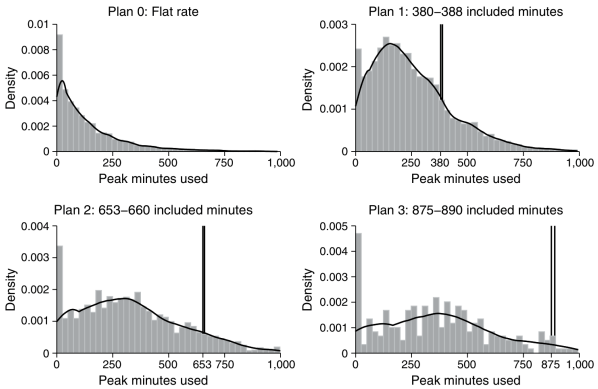


FIGURE 3. USAGE DENSITIES FOR POPULAR PLANS

Figure 3: Usage densities

7.3 Figures 4 and 5: inattention and model timing

Read:

- Figure 4 shows that calling increases when free periods begin even in months with usage well below allowance.
- This rules out a pure remaining-balance explanation and suggests consumers are inattentive to accumulated usage.
- Figure 5 summarizes the model timeline: signal, plan choice, threshold choice, taste realization, usage, and learning.

Economic message: Bill-shock alerts matter because consumers do not fully track their billing-cycle position. The model timing explains how information and learning enter decisions.

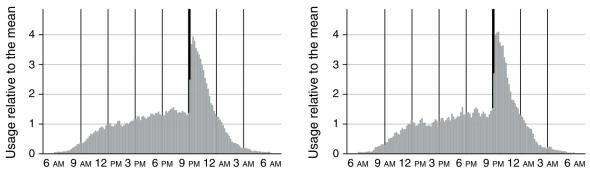


FIGURE 4. WEEKDAY USAGE PATTERNS IN MONTHS SUBSTANTIALLY BELOW ALLOWANCE

Notes: Weekday usage patterns of outgoing calls to landlines for plan 1–3 subscribers during months in which total usage was at most 65 percent of the included allowance. Usage patterns are shown for the first three weeks of the month (panel A) and the last week of the month (panel B).

Figure 4: Below-allowance usage patterns

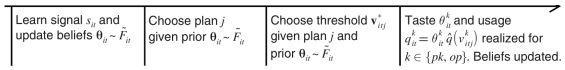


FIGURE 5. MODEL TIME LINE

Figure 5: Model timeline

7.4 Figures 6 and 7: calling threshold and belief identification

Read:

- Figure 6 shows the peak inverse demand curve and the optimal calling threshold.
- Calls with value above the threshold are made; calls below the threshold are not.
- Figure 7 shows how plan choice regions map into initial beliefs about mean and variance.
- The distribution of plan choices identifies the distribution of beliefs about future usage.

Economic message: The paper identifies beliefs from sorting across nonlinear plans and identifies calling behavior from within-period usage responses.

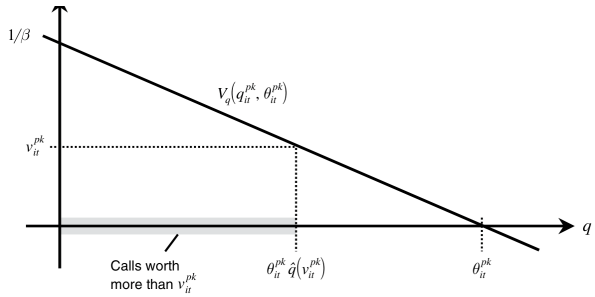


FIGURE 6. PEAK INVERSE DEMAND CURVE AND CALLING THRESHOLD

maximize her expected utility conditional on her period t informatic allowance Q_j, overage rate p_j, and multiplicative demand in equatic imal threshold derived in online Appendix B.1 is uniquely characteri

Figure 6: Inverse demand and threshold

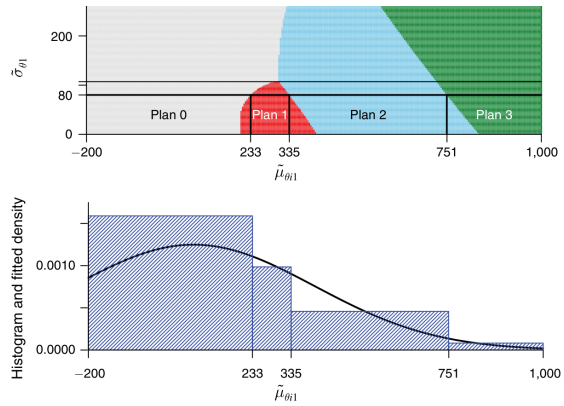


FIGURE 7. IDENTIFICATION OF BELIEFS

Notes: Top panel: Plan choice as a function of initial beliefs $\{\tilde{\mu}_{\theta i1}, \tilde{\sigma}_{\theta 1}\}$ implied by the model evaluated at October–November 2002 prices given $\beta = 2$. Bottom panel: Histogram and fitted normal distribution over $\tilde{\mu}_{\theta i1}$ implied by the assumption $\tilde{\sigma}_{\theta 1} = 80$ and October–November 2002 new subscriber plan choice shares of 69 percent, 10 percent, 19 percent, and 2 percent for plans 0 to 3, respectively.

Figure 7: Identification of beliefs

7.5 Tables 2 and 3: parameter estimates and bias estimates

Read:

- Table 2 reports structural parameters. Price sensitivity is estimated at $\beta = 2.698$.
- The overconfidence parameter is $\delta = 0.383$, implying that consumers substantially underestimate usage risk.
- Consumers' initial uncertainty and monthly volatility are large in true terms but understated in perceived terms.
- Table 3 summarizes the three biases: overconfidence, aggregate mean bias, and conditional mean bias.
- Aggregate mean bias is negative, and conditional mean bias is positive.

Economic message: The model estimates overconfidence, mean underestimation, and extreme-plan bias simultaneously, allowing the policy analysis to separate their welfare effects.

Parameter		Estimate	SE	Parameter		Estimate	SE
β	Price sensitivity	2.698	(0.487)	$\sigma_{\varepsilon}^{op}$	$SD[\varepsilon_{it}^{op}]$	325.473	(8.726)
$\tilde{\mu}_0^{pk}$	$E[\mu_{i1}^{pk}]$	216.398	(19.156)	ρ_{ε}	$\mathrm{corr}(\varepsilon_{it}^{pk}, \varepsilon_{it}^{op})$	0.517	(0.019)
μ_0^{pk}	$E[\mu_i^{pk}]$	271.799	(11.951)	$\rho_{s,pk}$	$\mathrm{corr}(s_{it}, \varepsilon_{it}^{pk})$	0.410	(0.022)
μ_0^{op}	$E[\mu_i^{op}]$	419.161	(32.404)	δ	Overconfidence	0.383	(0.017)
$\tilde{\sigma}_{\mu}^{pk}$	$SD[\mu_{i1}^{pk}]$	210.052	(24.130)	$\mu_{\alpha}^{pk,9}$	$E[\alpha_i^{pk,9}]$	−0.008	(0.002)
σ_{μ}^{pk}	$SD[\mu_i^{pk}]$	147.702	(9.321)	$\sigma_{\alpha}^{pk,9}$	$SD[\alpha_i^{pk,9}]$	0.061	(0.003)
σ_{μ}^{op}	$SD[\mu_i^{op}]$	421.077	(39.811)	$\sigma_{e}^{pk,9}$	$SD[e_{it}^{pk,9}]$	0.108	(0.003)
$\rho_{\tilde{\mu},pk}$	$\mathrm{corr}(\mu_{i1}^{pk}, \mu_i^{pk})$	0.195	(0.137)	$\sigma_{e}^{op,9}$	$SD[e_{it}^{op,9}]$	0.124	(0.003)
$\rho_{\tilde{\mu},op}$	$\mathrm{corr}(\mu_{i1}^{pk}, \mu_i^{op})$	0.012	(0.019)	P_C	Plan consideration	0.059	(0.003)
ρ_{μ}	$\mathrm{corr}(\mu_i^{pk}, \mu_i^{op})$	0.619	(0.075)	O	Outside good	1.159	(0.643)
$\sigma_{\varepsilon}^{pk}$	$SD[\varepsilon_{it}^{pk}]$	248.237	(10.948)				

Note: The log likelihood is −295,647.5.

Table 2: Parameter estimates

Bias	Parameter	Unbiased	Estimate	SE	Finding	Choose overly
Overconfidence	δ	$\delta = 1$	0.38	0.02	$\delta < 1$	risky plans
Aggregate mean bias	b_1	$b_1 = 0$	−55	19	$b_1 < 0$	small plans
Conditional mean bias	b_2	$b_2 = 0$	0.86	0.09	$b_2 > 0$	extreme plans

B. Biases and Learning

Table 3: Consumer biases

7.6 Tables 4 and 5: counterfactual welfare and bill-shock regulation

Read:

- Table 4 shows fixed-price surplus changes from eliminating biases.
- Removing overconfidence raises consumer welfare and lowers firm profits.

- Table 5 shows endogenous-price counterfactuals for bill-shock regulation and bias elimination.
- With prices fixed, bill-shock regulation raises annual consumer welfare by about 103.
- With endogenous prices, bill-shock regulation lowers annual consumer welfare by about 33 and total welfare by about 26.

Economic message: The policy lesson hinges on equilibrium pricing. A regulation that helps consumers at fixed prices can hurt them once firms adjust tariffs.

TABLE 4—ANNUAL PER-STUDENT CHANGE IN SURPLUSES FROM BIAS ELIMINATION (*Fixed prices*)

	University plans			Public plans		
	Profits	Consumer welfare	Total welfare	Profits	Consumer welfare	Total welfare
Welfare at estimates	261.64	1,154.54	1,416.18	321.41	1,083.94	1,405.35
$\delta = 1$	−53.39	51.11	−2.28	−88.96	75.87	−13.10
No biases	−94.24	90.55	−3.69	−78.66	90.92	12.27

Notes: Annual changes in surpluses (profits, consumer welfare, and total welfare) are measured in dollars per student averaged over the two-year sample period. Changes are relative to surpluses at estimates in row 1.

Table 4: Fixed-price bias counterfactuals

TABLE 5—EFFECT OF BILL-SHOCK REGULATION AND REMOVING BIASES WITH ENDOGENOUS PRICES

Biases:		Estimates	Estimates	Estimates	$\delta = 1$	No biases
Bill-shock regulation:		No	Yes (prices fixed)	Yes	No	No
		(1)	(2)	(3)	(4)	(5)
Plan 1	M	42.88	42.88	39.28	42.32	52.59
	Q	216	216	0	0	0
	p	0.50	0.50	0.17	0.13	0.07
	Share	39	43	26	42	37
Plan 2	M	48.64	48.64	50.66	70.63	69.41
	Q	383	383	80	∞	∞
	p	0.50	0.50	0.12	N/A	N/A
	Share	38	36	23	46	52
Plan 3	M	58.12	58.12	68.23		
	Q	623	623	540		
	p	0.50	0.50	0.12		
	Share	14	11	40		
Outside good share		10	10	12	11	11
Usage		240	199	239	262	288
Overage revenue		223	2	152	136	75
Annual profit		501	305	509	512	512
Annual consumer welfare		903	1,006	870	884	907
Annual total welfare		1,404	1,311	1,379	1,396	1,419
Δ annual profit			−196	7	11	11
Δ annual consumer welfare			103	−33	−19	4
Δ annual total welfare			−93	−26	−8	15

Notes: All welfare and profit numbers are expressed in dollars per customer per year. Because the counterfactuals in columns 4 and 5 produced two-part tariffs, bill-shock regulation has no additional effect. We simulate 10,000 consumers for 12 months.

C. *Counterfactual Simulation Results*

Table 5: Endogenous-price regulation counterfactuals

7.7 Figure 8: distributional consequences

Read:

- The left panel shows the distribution of annual utility changes from bill-shock regulation.
- Some consumers benefit substantially from avoiding large overage bills.
- The right panel shows that bill-shock regulation shifts overage fees downward, especially reducing large overages.

Economic message: The average welfare effect is negative, but the regulation has an insurance-like distributional benefit for consumers who would otherwise experience large bills.

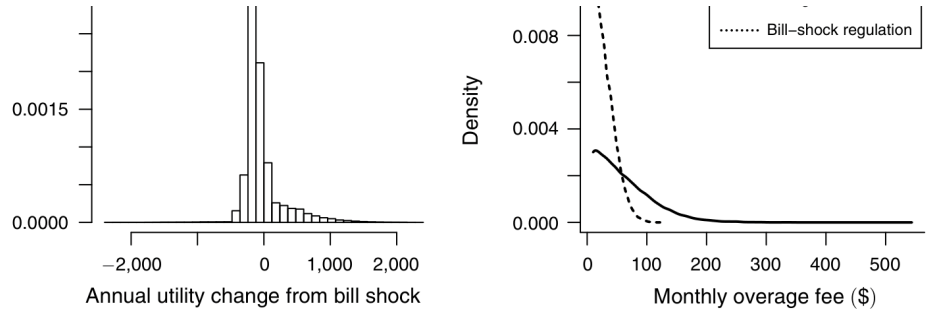


FIGURE 8. DISTRIBUTIONAL CONSEQUENCES OF BILL-SHOCK REGULATION

es: Left panel: Histogram of changes in consumer utility due to bill-shock regulation. Right panel: Density of rage fees with and without bill-shock regulation (excluding atoms at zero).

8 Short summary

- The paper studies cellular plan choice using detailed billing data from 2002–2004.
- Consumers choose nonlinear tariffs before knowing future usage.
- Usage responds sharply to marginal prices, but consumers do not perfectly forecast or monitor usage.
- The structural model includes plan choice, calling thresholds, learning, and biased beliefs.
- The paper estimates substantial overconfidence: consumers underestimate future usage variance.
- Consumers also have aggregate and conditional mean biases.
- At fixed prices, removing biases helps consumers and reduces firm profits.
- Bill-shock alerts look beneficial at fixed prices.
- After equilibrium price adjustment, bill-shock regulation lowers average consumer welfare.
- The core lesson is that consumer-protection regulation must account for firm price responses.

9 Conclusion

9.1 Core conclusion

Grubb and Osborne show that consumer mistakes in nonlinear tariff markets matter quantitatively, but policy conclusions depend critically on firm pricing responses. Consumers in the sample are inattentive and overconfident, and these biases affect plan choice, usage, and overage charges.

9.2 Economic intuition

Nonlinear tariffs exploit differences in risk and attention. Overconfident consumers choose plans that expose them to high overage risk. Bill-shock alerts reduce the surprise component of overage payments, but firms respond by changing fixed fees, allowances, and overage prices. The final welfare effect depends on both consumer behavior and equilibrium tariff design.

9.3 Incremental contribution in one sentence

The paper's incremental contribution is to combine detailed usage data, a behavioral structural model of plan choice and learning, and endogenous price counterfactuals to show that a seemingly protective information

regulation can reduce consumer welfare once firms redesign nonlinear tariffs.

9.4 Takeaway

The broader takeaway is that behavioral market failures and industrial organization cannot be analysed separately. Correcting consumer mistakes changes firm incentives, and those price responses can dominate the direct benefits of the correction.